DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADESH, LUCKNOW

EVALUATION SCHEME FOR & SYLLABUS

B. TECH. FOURTH YEAR

Electronics Engineering
(VLSI Design and Technology)

AS PER

AICTE MODEL CURRICULUM & NEP2020

[Effective from the Session: 2026-27]

B.Tech. VII Semester

B. Tech - Electronics Engineering (VLSI Design and Technology)

Sr. No.	Course Code	Course Title	Learning Mode	Periods			Evaluation Scheme				End Semester		Total	Credit
				L	T	P	CT	TA	Total	PS	TE	PE		
1	BEC701	VLSI Design	Offline	3	0	0	20	10	30	-	70	-	100	3
2	BEC07X	Departmental Elective-IV	Offline	3	0	0	20	10	30	-	70	-	100	3
3	BOE**	Open Elective-II (M)	Offline	3	0	0	20	10	30	-	70	-	100	3
4	BEC751	VLSI Design Lab	Offline	0	0	2	-	-	-	50	-	50	100	1
5	BEC752	Mini Project or Internship Assessment*		0	0	4	-	-	-	100	-	-	100	2
6	BEC753	Project-I		0	0	10	-	-	-	150	-	-	150	5
7	BEC754	Start-up and Entrepreneurial Activity Assessment#		0	0	4	-	-	-	100	-	-	100	2
		Total		**9**	**0**	**20**							**750**	**19**
		PC- Program Core *The Mini Project or internship (5 - 6 weeks) conducted during summer break after VI semester and will be assessed during VII semester. #The Startup and Entrepreneurial Activity Assessment will be done in 7th semester under which a student will have to undergo a startup/entrepreneurship activity of at least 60 hours till 6th semester												

Course Code	Course Title Department Elective-IV
BEC-074	Drone Systems: Design, Control & Navigation
BEC-076	ASIC Design and Verification
BEC-077	Mixed Signal Circuits
BEC-078	VLSI Design Flow: RTL to GDS

B.Tech. VIII Semester

B. Tech - Electronics Engineering (VLSI Design and Technology)

Sr. No.	Subject Code	Subject	Learning Mode	Periods			Evaluation Scheme				End Semester		Total	Credits
				L	T	P	CT	TA	Total	PS	TE	PE		
1	BOE**	Open Elective-III	Offline	3	0	0	20	10	30	-	70	-	100	3
2	BOE**	Open Elective-IV	Offline	3	0	0	20	10	30	-	70	-	100	3
3	BEC851	Project-II		0	0	18	-	-	-	100	-	350	450	10
		Total		6	0	18							650	16

B.Tech 4rd Year
VII Semester
Syllabus

BEC-701	VLSI Design	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Introduction:** VLSI Design flow, general design methodologies; critical path and worst-case timing analysis, overview of design hierarchy, layers of abstraction, integration density and Moore's law, VLSI design styles, packaging, CMOS Logic, Propagation Delay definitions, sheet resistance.	8
II	**Interconnect Parameters:** Resistance, Inductance, and Capacitance, skin effect and its influence, lumped RC Model, the distributed RC Model, transient Response, RC delay model, Linear Delay Model, Logical Effort of Paths, Scaling.	8
III	**Dynamic CMOS design:** steady-state behavior of dynamic gate circuits, noise considerations in dynamic design, charge sharing, cascading dynamic gates, domino logic, np-CMOS logic, problems in single-phase clocking, two- phase non-overlapping clocking scheme, Sequential CMOS Logic Circuits, Layout design.	8
IV	**Semiconductor Memories:** Dynamic Random Access Memories (DRAM), Static RAM, non-volatile memories, flash memories, Pipeline Architecture. Low – Power CMOS Logic Circuits: Introduction, Overview of Power Consumption, Low – Power Design through voltage scaling,	8
V	**Introduction to Testing:** Faults in digital circuits. Modeling of faults, Functional Modeling at the Logic Level, Functional Modeling at the Register, Structural Model and Level of Modeling. Design for Testability, Ad Hoc Design for Testability Techniques, Controllability and Observability, Introduction to Built-in-self-test (BIST) Concept.	8

Text Book:

1. Sung-Mo Kang & Yosuf Leblebici, "CMOS Digital Integrated Circuits: Analysis & Design",Mcgraw Hill, 4th Edition.
2. Neil H.E.Weste, David Money Harris, "CMOS VLSI Design – A circuits and Systems Perspective" Pearson, 4th Edition.
3. D. A. Pucknell and K. Eshraghian, "Basic VLSI Design: Systems and Circuits", PHI, 3rd Ed.,1994.

Reference Books:

1. R. J. Baker, H. W. Li, and D. E. Boyce , " CMOS circuit design, layout, and simulation", Wiley-IEEE Press, 2007.
2. M. Abramovici, M.A. Breuer and A.D. Friedman, "Digital Systems and Testable Design" , Jaico Publishing House.

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Express the concept of VLSI design and CMOS circuits and delay study.
2. Analyze mathematical methods and circuit analysis models in analysis of CMOS digital electronics circuits.
3. Design and analyze various combinational & sequential circuits based

on CMOS technology.

4. Examine power logic circuits and different semiconductor memories used in present day technology.
5. Interpret faults in digital circuits, Fault Models and various Testing Methodologies.

BEC-751	VLSI Design Lab	0L:0T:2P	1 Credit

SUGGESTIVE LIST OF EXPERIMENTS:

1. Design and analysis of basic of logic Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR.
2. Design and implementation of Half adder and Full adder using CMOS logic.
3. To simulate the schematic of the common drain amplifier.
4. To simulate the schematic of the differential amplifier.
5. To simulate the schematic of the operational amplifier.
6. Design of 3-8 decoder using MOS technology.
7. Design a 4:1 Multiplexer.
8. Design and implementation of Flip flop circuit.
9. Layout design of PMOS, NMOS transistors.
10. Layout design of CMOS inverter and its analysis.

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Designing of logic gates.
2. Implementation of combinational and sequential circuits using CMOS logic.
3. Analyze amplifier circuits.
4. Design sequential circuits such as flip flop.
5. Do the layout designing for physical analysis of the MOS transistor and MOS based circuits.

BEC-074	Drone Systems: Design, Control & Navigation	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**UAV Miniaturization:** Design Challenges and Technological Opportunities, Aerial Network Characteristics: Dynamic Topology, Delay Constraints, Intermittent Links, Aerial Communication Techniques: Radar Identification & Limitations Beyond Radar: ADS-B, Optical & Acoustic Tracking, Aerial Voice & Data Communication.	8
II	**Modern Control Theory for UAV Systems:** Introduction to State-Space Modeling, State Transition Matrix and Time Response, Canonical Forms & Transformations, Significance of Eigenvalues: Real Distinct, Repeated, Complex, Application to UAV Stability & Control Systems.	8
III	**UAV Aerodynamics, Propulsion & Navigation:** Coordinate Systems: Earth-fixed & Body-fixed Frames, UAV Aerodynamics: Wing Configuration, Lift & Drag, Airframe Configuration & Endurance, Vibration, Noise, and Gliding Characteristics; UAV Propulsion: Propeller & Motor Modeling, Electric vs. Hybrid Propulsion Systems.	8
IV	**UAV Control Systems & Autonomy:** UAV Controllability & Observability Concepts, UAV Flight Control System Overview, Position Control: Euler Angles and Rotation Matrix as Outputs, Control Strategies: Autonomous vs. Semi-Autonomous Flight, Inner-loop vs. Outer-loop Control, Safety Mechanisms: Failsafe Design, Redundancy and Recovery in Autonomous Flights.	8
V	**Drone Safety, Legal Framework & Future Trends:** National and International Safety Guidelines: DGCA Drone Rules 2021 (India), Airspace Classification & Permissions, Pilot Training, UIN, UAS Traffic Management (UTM), NPNT Compliance & Geo-fencing Technologies, Innovations in Drone Miniaturization & Battery Technologies.	8

Text Books:

1. Reg Austin, Unmanned Aircraft Systems: UAV Design, Development and Deployment, Wiley, 2010.
2. Robert C. Nelson, Flight Stability and Automatic Control, McGraw-Hill, 2nd Edition, 1998.
3. Brian L. Stevens, Frank L. Lewis, Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems, Wiley, 3rd Edition, 2015.

Reference Books:

1. Paul G. Fahlstrom, Thomas J. Gleason, Introduction to UAV Systems, UAV Systems Inc., 4th Edition, 2012.
2. Andreas M. Savvaris, Kostas P. Valavanis, Unmanned Aircraft Systems: Theory and Practice, Springer, 2020.
3. DGCA (Directorate General of Civil Aviation, India), Drone Rules 2021 – Government of India (available online).

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Understand the constraints and opportunities in UAV miniaturization and aerial networking.
2. Understand how to represent drone systems using state-space models and use basic control theory to analyze their stability.
3. Analyze the aerodynamic behaviour and propulsion trade-offs in UAV design and navigation.
4. Recognize UAV control systems using modern control techniques for both autonomous and semi-autonomous operations.
5. Identify the rules and safety practices for drone operation in India and globally, including licensing, geofencing, and UAS traffic management.

BEC-076	ASIC Design and Verification	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**ASIC Design Flow and System Verilog Fundamentals** Overview of ASIC design flow, Role of RTL and Verification in ASIC design. SystemVerilog overview and evolution from Verilog. Data Types: logic, bit, reg, wire, Constants, Net Type, Arrays: static, dynamic, associative, queues, Enumerated types, Typedef and packages, String, Operators and expressions. Procedural statements and Routines - Procedural Statements, Specialized constructs: always_comb, always_ff, always_latch. Tasks, Functions, and Void Functions, Routine arguments – Returning from a routine.	8
II	**Combinational Circuit Modeling using SystemVerilog** Continuous assignments. Sensitivity lists and always_comb., Blocking assignment, Modeling and verification of-Multiplexer, Encoder / Decoder, Adder / Subtractor, Comparator, Parameterized combinational modules	8
III	**Sequential Circuit Modeling using SystemVerilog** always_ff construct, non-blocking assignment, Modeling and verification of-Flip-flops, Counters (Up/Down, Mod-N), Shift registers, FSM design (Moore & Mealy). Clock and reset strategies (synchronous/asynchronous).	8
IV	**Verification Fundamentals** Need for verification, Testing Vs Verification, Separation of Testbench and Design, Testbench architecture. Interface Construct, Stimulus Timing, Module interactions. Direct Testing. Self-checking testbench. Scoreboard concept. System Verilog Assertions, Procedural and Concurrent Assertions.	8
V	**Digital Design using FPGA** FPGA Design flow, Applications of FPGA, FPGA Architecture, Introduction to FPGA development environments (Vivado Tool), FPGA implementation of combinational circuits: logic gates, adder, subtractor, decoder, multiplexer. FPGA implementation of sequential circuits: Flip flops and counter with & without inclusion of constraint file.	8

Text Books:

1. Christian B Spear, “System Verilog for Verification: A guide to learning the Testbench language features”, Springer publications, Third Edition, 2012.
2. Stuart Sutherland, Simon Davidmann and Peter Flake, SystemVerilog for Design: A Guide to Using SystemVerilog for Hardware Design and Modeling, Springer, 2nd Edition, 2006.
3. Z. Navabi, Digital System Design with FPGA: Implementation Using Verilog and VHDL, Springer International Publishing, 2018.

Reference Books:

1. Ashok B. Mehta, Introduction to SystemVerilog, Springer, 2nd Edition, 2021.
2. Sanjay Churiwala, Usha Mehta and Vaishali Dhare, Digital Design and Verification using Verilog and SystemVerilog, Wiley India, 2024.
3. Mark Zwoliński, Digital System Design with SystemVerilog, Prentice Hall Modern Semiconductor Design, 2010/2016.
4. Janick Bergeron, Writing Testbenches using SystemVerilog, Synopsys Inc., Springer Publications, 2006.

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Understand ASIC design flow and SystemVerilog fundamentals.
2. Apply SystemVerilog constructs to implement combinational circuits.

3. Apply SystemVerilog constructs to implement sequential circuits.
4. Analyze structured testbenches using assertions and coverage techniques.
5. Construct Digital Circuits on FPGA platform.

BEC-077	Mixed Signal Circuits	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Basic Analog Building Blocks** Review of MOSFET model, Differential amplifier (CMOS), Ideal operational amplifier model, Practical operational amplifier model, Non-ideal parameters: gain, bandwidth, slew rate, offset, Inverting and non-inverting configurations, Op-amp as comparator	8
II	**Digital-to-Analog Converter Fundamentals and Architectures** Data Converter fundamentals: Analog versus Digital Discrete Time Signals, Digital-to-Analog Converter (DAC): Introduction, specifications: Resolution, Full scale voltage, Least Significant Bit (LSB), Most Significant Bit (MSB), percentage accuracy, Differential Non Linearity (DNL), Integral Non-Linearity (INL), offset, gain error, Latency, Signal to Noise Ratio (SNR), Dynamic Range. DAC Architectures: Resistors String, R-2R Ladder Networks, Current Steering, Introduction to charge scaling DAC	8
III	**Analog-to-Digital Converter Fundamentals and Architectures** Analog-to-Digital Converter (ADC): Introduction, Specifications: Quantization, Quantization error, Differential non-linearity, Missing codes, Integral Non-linearity, Offset and gain error, SNR, Aperture error, ADC Architectures: Flash, 2-step Flash ADC, Pipeline ADC, Integrating ADC, Successive Approximation ADC.	8
IV	**Sample & Hold and Switched Capacitor Circuits** Introduction- Performance of sample and hold circuits, Testing Sample and Holds, MOS sample and Hold Basics, Example of CMOS sample and Hold Circuits, Basic operation and analysis of switched capacitor circuits, resistor equivalence of a switched capacitor, noise in switched-capacitor circuits.	8
V	**RF and Mixed-Signal Integrated Circuits** Introduction to RF Integrated Circuits, RF Performance Parameters – Gain, Bandwidth, Noise Figure (NF), Dynamic Range, Signal-to-Noise Ratio (SNR), Signal-to-Interference Ratio (SIR), Spurious-Free Dynamic Range (SFDR), RF Passive Components, Low Noise Amplifier (LNA), RF Mixers	8

Text Books:

1. B. Razavi, “Principles of Data Conversion System Design”, 1st Edition, Wiley-IEEE Press, 1994
2. R. J. Baker, “CMOS Mixed Signal circuit Design”, 2nd Edition, Wiley 2008.

Reference Books:

1. P.E. Allen and D.R. Holberg, “CMOS Analog Circuit Design”, Oxford University Press, 2nd Edition, 2002.
2. D. P. Kothari and J. S Dhillon, “Digital Circuits and Design”, Pearson, 2016.
3. Charles H Roth, “Fundamentals of Logic Design”, Cengage Learning.

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Explain MOSFET-based analog building blocks and op-amp non-ideal characteristics.
2. Evaluate DAC performance parameters (DNL, INL, SNR, accuracy) and compare different DAC architectures.
3. Examine ADC specifications and select suitable architectures for given design requirements.

4. Interpret the operation of sample-and-hold and switched-capacitor circuits considering noise and performance limitations.
5. Examine RF and Mixed-Signal Integrated Circuits.

BEC-078	VLSI Design Flow: RTL to GDS	3L:0T:0P	3 Credits

Unit	Topics	Lectures
I	**Fundamentals of VLSI Design Flow and HDL** Basic concept of Integrated Circuits, Moore's law, Scales of Integration, Introduction to ASIC and SoC design, VLSI Design flow, FPGA Design Flow, Technology nodes, GDSII file format definition, PDK definition, Introduction to EDA tools used in front-end and back-end design. VLSI Design Styles Basics of Hardware Description Languages (HDLs): Verilog HDL fundamentals, modeling styles – Dataflow, Behavioral and Structural modeling. Module definition, ports, data types and operators in verilog.	10
II	**RTL Design and Functional Verification** Blocking vs Non-blocking assignments, Always blocks-combinational & sequential modeling, Control blocks: if-else, case and loops. RTL implementation of combinational and sequential circuits. FSM design (Moore & Mealy). Functional simulation and test benches development. Introduction to simulation tools (e.g., Xcelium) and waveform analysis.	8
III	**Logic Synthesis and Static Timing Analysis** Logic synthesis flow: RTL to gate-level netlist. Technology mapping and synthesis constraints (introduction to SDC). Inputs and outputs of synthesis, Gate-level netlist generation and analysis. Overview of synthesis optimization (timing, area and power – introductory). Synthesis tools overview. Static Timing Analysis (STA): Propagation delay, clock skew, and clock jitter. Setup and hold check. Timing paths, slacks and timing reports.	7
IV	**Introduction to Physical Design** Overview of Physical Design flow. Floorplanning concepts and partitioning (basic). Placement of standard cells and macros (overview). Clock Tree Synthesis (CTS) – concept and objectives. Routing – global and detailed routing (introductory). Power planning basics (power rings, power stripes). Congestion analysis (basic understanding).	7
V	**Physical Verification and Signoff** Design Rule Checking (DRC) – purpose and methodology. Layout vs Schematic (LVS), Parasitic extraction (RC extraction overview), Introduction to Signal integrity and IR drop concepts, Tape-out procedure and GDSII generation.	8

Text Books:

1. Sneh Saurabh, Introduction to VLSI Design Flow, McGraw Hill Education (India), 1st Edition, 2018.
2. Samir Palnitkar, Verilog HDL: A Guide to Digital Design and Synthesis, Pearson Education, 2nd Edition, 2003.
3. Naveed A. Sherwani, Algorithms for VLSI Physical Design Automation, Springer, 3rd Edition, 1999.

Reference Books:

1. Michael D. Ciletti, Advanced Digital Design with the Verilog HDL, Pearson Education, 2nd Edition, 2011.
2. Khosrow Golshan, Physical Design Essentials: An ASIC Design Implementation Perspective, Springer, 1st Edition, 2007.
3. S. K. Lim, Practical Problems in VLSI Physical Design Automation, Springer, 1st Edition, 2013.

Course Outcomes: At the end of this course students will demonstrate the ability to:

1. Explain the complete VLSI design flow from specification to tape-out.
2. Design and simulate RTL modules using Verilog HDL.
3. Perform logic synthesis and analyze timing, area, and power reports.

4. Explain basic physical design stages including floor planning, placement, CTS and routing.
5. Perform physical verification and interpret signoff result.